

IMPACTS OF PIPELINE-LEVEL DIFFERENCES ON ATMOSPHERIC RETRIEVALS



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Department of Astrophysics

S. Schleich¹, S. Boro Saikia¹, Q. Changeat^{2,3}, M. Güdel¹, A. Voigt⁴, I. Waldmann³

CONTEXT

- Observational characterisation of exoplanet atmospheres via **transmission spectroscopy**
- Inversion techniques (*atmospheric retrieval*) **depend on the underlying data set**

OBJECTIVE

What is the impact of assumptions made during data reduction on parameter inference?

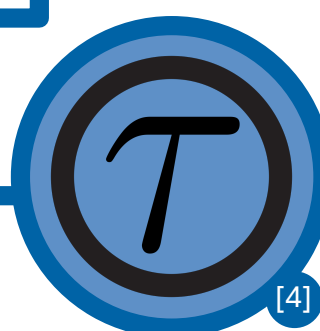
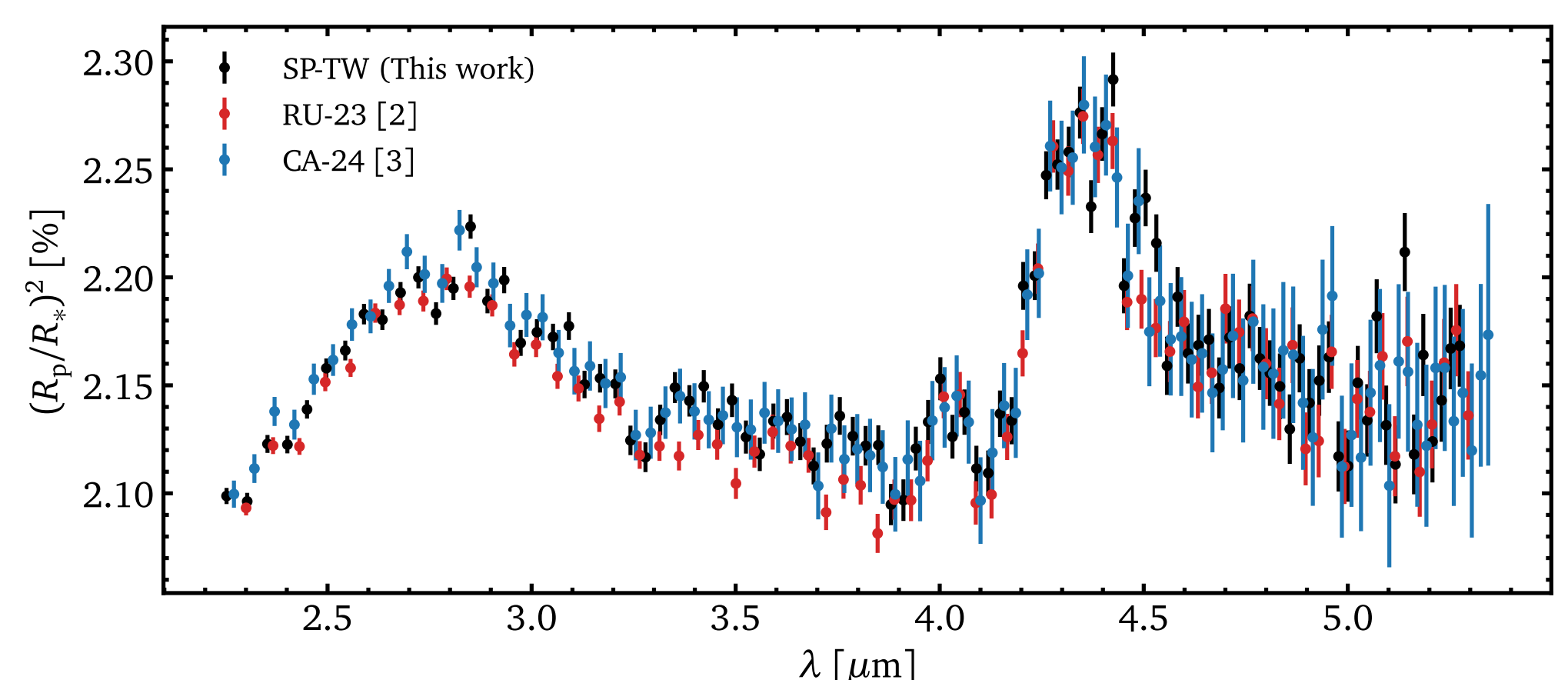
Homogenised atmospheric retrievals on three spectra derived from the same observation, using the same pipeline

Data reduction for **WASP-39 b**
(JWST NIRSpec PRISM)



THE DATA

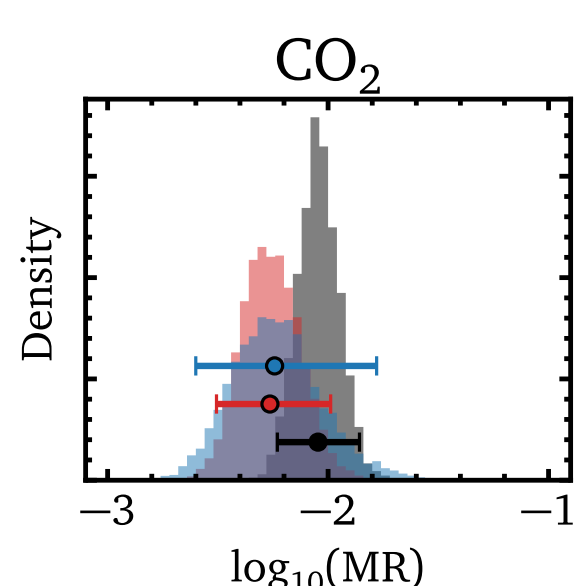
- Comparison the established data products from the same pipeline [2, 3] shows **systematic differences**



Atmospheric retrievals using *TauREx3*

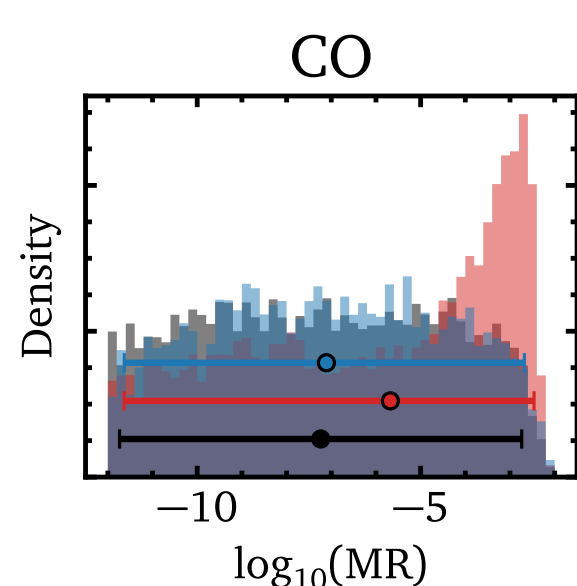
CHARACTERISATION RESULTS

- Homogenised atmospheric retrievals tuned on SP-TW
- We find **three classes** of posterior distributions:



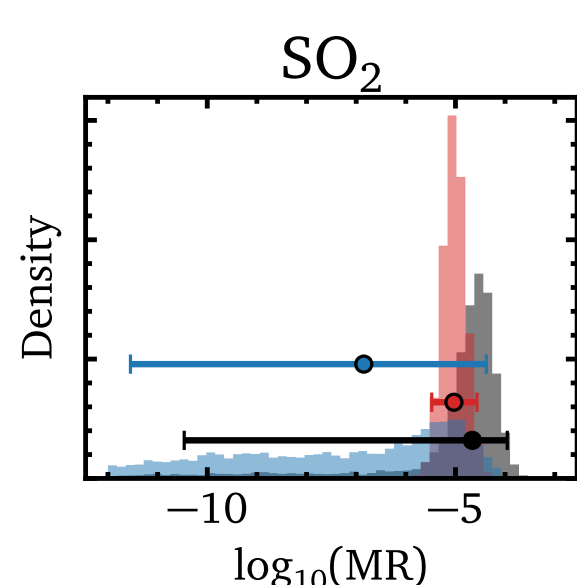
Stable and Gaussian

Constituents with absorption features over large wavelength ranges



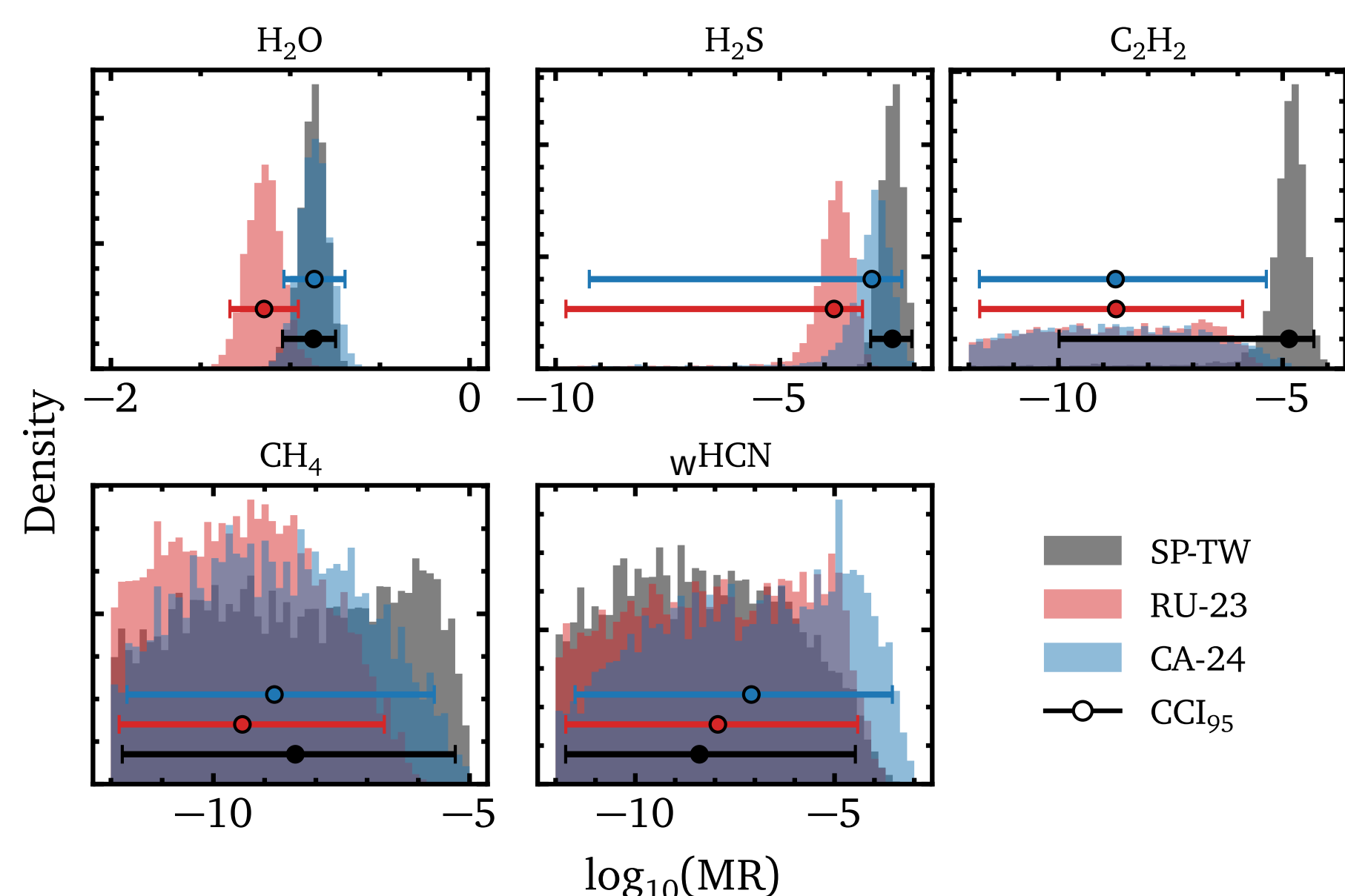
Stable and uniform

Constituents with no clear absorption signatures



Unstable and asymmetric

Constituents retrieved over only a small wavelength region



CONCLUSIONS

- (1) Results of atmospheric retrievals are **sensitive to small-scale variations** in the inference-guiding, high-precision data.
- (2) Population-level parameter investigations will require **homogenised data reduction** efforts.
- (3) Small credible intervals (e.g., '1σ') can provide a **misleading sense of accuracy**

Affiliations:

1. Department of Astrophysics, *University of Vienna*
2. Kapteyn Astronomical Institute, *University of Groningen*
3. Department of Physics and Astronomy, *University College London*
4. Department of Meteorology and Geophysics, *University of Vienna*

Acknowledgements:

S. Schleich acknowledges funding from the *University of Vienna* and the *Austrian Research Association (ÖFG)*. The computational results of this work have been achieved in part using the *Austrian Scientific Computing (ASC) infrastructure*.

Check out the paper:
A&A, 704, A223



References:

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simon.schleich@univie.ac.at